SubMind

MIS 321 PHASE 1

Preliminary Investigation Report

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INTRODUCTION

In today's digital age, almost everyone has more than one subscription. People watch movies on Netflix, listen to music on Spotify, go to the gym, or spend time on gaming platforms. However, managing all these subscriptions at the same time is often difficult for users. Forgetting payment dates, not noticing unused memberships, or continuing to pay after a free trial has ended are some of the most common problems.

SubMind is designed to solve exactly these problems. The application aims to make life easier by bringing all subscriptions together in one place. For young people, students, and busy professionals, saving both time and money has become an important need.

Although there are some apps in the market for tracking subscriptions, in Turkey there is no comprehensive solution that works fully integrated with banks, App Store/Google Play, and the e-invoice system. SubMind aims to fill this gap and become a practical and secure system that users really need.

The importance of the system is not only about listing subscriptions. With features such as reminder notifications, free-trial ending alerts, detecting unused subscriptions, and monthly expense reports, SubMind aims to be a tool that not only makes daily life easier but also helps users increase their financial awareness.

SWOT ANALYSIS

STRENGTHS

- Designed specifically for the Turkish market (bank, e-invoice, App Store/Google Play integrations)
- Secure data management fully compliant with KVKK and GDPR
- Detection of unused subscriptions and price increase alerts
- Clear value proposition for young professionals and students: saving money + convenience

WEAKNESSES

- New application → low initial brand awareness
- Integrations may be costly and time-consuming
- Risk of launching with limited features in the MVP version
- Potential user concerns regarding data sharing and privacy

OPPORTUNITIES

- Early advantage through open banking integrations
- Growing market due to the increasing number of subscription-based services
- High potential for adoption among university students and young professionals
- Long-term potential to evolve into a personal finance management tool

THREATS

- Entry of global competitors (e.g., Rocket Money, Bobby) into the Turkish market
- Possible regulatory changes (data privacy or payment system laws)
- Users preferring Apple/Google's built-in subscription management tools
- Banks and fintech companies developing similar features within their own apps

COMPETITOR ANALYSIS

The subscription management market has been growing rapidly worldwide, and there are several applications designed to help users manage their subscriptions. The most well-known global examples include **Rocket Money (formerly Truebill)**, **Bobby**, and **TrackMySubs**, while in Turkey, **Abonesepeti**, **SubX**, and **Abonelik Yöneticisi** have recently entered the market.

Rocket Money (Truebill) is widely used in the United States. It offers advanced features such as automatic subscription detection through bank account analysis, upcoming payment reminders, and even concierge cancellation services. However, due to differences in financial infrastructure and open banking regulations, it cannot operate at full capacity in Turkey.

Bobby is a simple and elegant subscription tracking app, especially popular among iOS users. It allows manual subscription entry and provides payment reminders, but lacks financial integrations, price comparisons, and smart analytics.

TrackMySubs is mainly aimed at freelancers and small businesses. It includes calendar views, CSV import/export, and expense tracking, but it may be too complex for regular individual users.

In Turkey, some local apps such as **Abonesepeti** and **SubX** have started to appear. These apps help users list and track their subscriptions but lack advanced features such as bank integration, e-invoice connection, price comparison, and automated cancellation support. Moreover, most local solutions do not clearly address data privacy and KVKK compliance, which are crucial in the Turkish market.

SubMind differentiates itself by being designed specifically for Turkish users. The system aims to:

- Integrate banking infrastructure, App Store/Google Play, and the national e-invoice system,
- Detect unused or rarely used subscriptions,
- Compare prices between platforms and suggest more affordable alternatives,
- Provide reminder notifications, free-trial alerts, and price increase warnings,
- And ensure secure data management fully compliant with KVKK and GDPR regulations.

Competitor Comparision Table

Feature / Platform	Rocket Money (Truebill)	Bobby	TrackMySubs	Abonesepeti / SubX (Turkey)	SubMind (Proposed)
Automatic Subscription Detection	Yes (via bank accounts)	No (manual input only)	Partial (CSV import/export)	No	Yes (via Turkish banks & open banking)
Payment Reminders	Yes	Yes	Yes	Yes	Advanced (payment, trial, price alerts)
Cancellation Support	Concierge cancellation	Not available	Manual only	Limited	Local cancellation guides & automation
Price Comparison	Limited	No	No	No	Yes (suggests cheaper alternatives)
Bank / E-Invoice Integration	U.S. banks only	No	No	Partial	Full Turkish banking & e-invoice integration
Data Privacy Compliance	GDPR / U.S. laws	Basic privacy policy	Basic	Limited	Full KVKK + GDPR compliance
Target Market	U.S. consumers	iOS users	Freelancers & SMEs	Turkish users	Turkish users, students & professionals
Value Proposition	Subscription automation & savings	Simple tracking	Business expense control	Basic tracking	Smart, localized, cost-saving system

In short, while existing applications mostly allow users to only view or track their subscriptions, **SubMind** takes a more comprehensive approach. It helps users not just manage their subscriptions but also save money, stay in control, and increase financial awareness.

By combining local integration, intelligent automation, and a user-focused design, SubMind aims to fill a major gap in the Turkish market as a truly smart and localized subscription management system.

1. **Subscription Management:**

Users can manually add or automatically import subscriptions from various platforms (Netflix, Spotify, Amazon, Gym, etc.).

All subscriptions are displayed in a single dashboard with details such as payment amount, due date, and renewal status.

2. **Reminders and Notifications:**

Automatic alerts for upcoming payments and expiring free trials.

Notifications for price changes or subscription renewals.

3. **Analytics and Reporting:**

Monthly spending reports showing total subscription costs.

Detection of rarely used or inactive subscriptions.

Visual charts for better financial awareness and budget control.

4. **Price Comparison and Recommendations:**

Comparison between similar platforms (e.g., Netflix vs. Disney+, Spotify vs. YouTube Music).

Personalized suggestions for cheaper or more suitable alternatives.

5. **Integrations:**

Bank integration for payment tracking through open banking.

Connection with App Store / Google Play to import mobile subscriptions.

E-invoice integration for official payment verification.

6. **User-Friendly and Secure Design:**

Mobile (iOS/Android) and web-based interfaces.

Two-factor authentication and encrypted data storage.

Full compliance with KVKK and GDPR regulations.

SubMind will save users both time and money by providing a centralized, intelligent, and secure solution for subscription management. It will also increase users' financial awareness by tracking their digital spending habits in an accessible and transparent way.

PRELIMINARY INVESTIGATION FINDINGS

Interview

As part of the preliminary research for our SubMind project, we conducted two separate interviews — one focusing on the business side and the other on the technical side.

The first interview was held with a CRM Manager working in the construction sector. This interview focused on user behavior, market expectations, trust factors, and identifying suitable revenue models for subscription management applications in Turkey.

The second interview was conducted with a mobile app developer. This discussion focused on the technical requirements, development challenges, and infrastructure preferences necessary for building a reliable and scalable mobile application.

These two perspectives provided complementary insights for our project. The business expert contributed to shaping the value proposition and market strategy, while the technical expert helped define the system's functionalities, integrations, and security priorities that will guide the development process.

Interview 1 – Business Perspective

Interview Purpose:

The purpose of this interview is to understand user needs, market dynamics, and the strategic factors required for the success of a subscription management system in Turkey. The goal is to gain insights into key topics such as user experience, security, revenue models, and customer retention strategies.

Interview Information:

Interviewee: Veli Asım Öztürk – CRM Manager

Organization: Mesa Holding

Date: October 2025

Method: Written interview (via Google Forms)

Duration: Approximately 25 minutes

Questions Asked:

1. What essential features should a subscription management system have to provide real value to users?
2. From a user experience perspective, which is more important in subscription management applications — design simplicity or advanced features?
3. In what situations do Turkish users feel the greatest need for such an application?
4. What methods are most effective in increasing user loyalty?
5. In your opinion, which aspects of such an application should be transparent to build user trust?
6. What would be the most suitable revenue model for such an application in the Turkish market?
7. Under what conditions would users be more willing to pay for such an application?
8. Do you think a subscription management system would be more successful as a standalone application or as a feature integrated into banking/fintech platforms?
9. Which features should be continuously improved to ensure users keep paying for the service over time?
10. What strategic steps are most critical for such an application to succeed in the Turkish market?
11. How important are business partnerships (banks, telecom companies, content providers) for the growth of such an application?
12. What is the most effective strategy to build user trust?

Interview Results:

The interview findings indicate that users place great importance on multi-platform support, security, and ease of use in subscription management systems. According to the interviewee, a successful system should not only work with popular platforms but also integrate with different sources such as utility bills and niche subscription services.

From a user experience standpoint, simplicity and smooth navigation should be prioritized. Although advanced features are valuable, they can make the interface complicated, leading to a poorer user experience. Therefore, a clear and user-friendly interface is essential.

Users mostly need such an application when they face difficulties managing multiple payments and subscriptions with different due dates. In this context, the system should help users manage their budgets and prevent unnecessary expenses, offering real financial value.

To increase user engagement, small incentives such as discounts, reports, and timely notifications were suggested. In terms of security, two-factor authentication (2FA) and certifications from official authorities were highlighted as must-have features.

For the revenue model, the freemium/premium structure was recommended. For example, free users could link a limited number of accounts, while premium members could enjoy unlimited access and advanced reporting features.

In the long term, partnerships with public institutions, banks, and telecom companies were seen as essential for sustainability. Such collaborations would not only increase trust but also expand the application's coverage and functionality.

Finally, the most effective strategy for building trust was identified as obtaining official certifications and running marketing campaigns with reliable financial influencers on social media.

Interview 2 – Technical Perspective

Interview Purpose:

The purpose of this interview is to identify the technical requirements of the SubMind system and to understand which technologies and infrastructure approaches would be most suitable for building a secure and scalable subscription management platform. The discussion mainly focused on the selection of mobile frameworks, backend structures, databases, and key security practices.

Interview Information:

Interviewee: Adar Keser – Mobile App Developer

Organization: Freelancer

Date: October 2025

Method: Written interview (via WhatsApp)

Duration: Approximately 15 minutes

Questions Asked:

1. What kind of technology stack (mobile framework, backend language, database) would you prefer for developing a subscription management application?

2. As the number of users increases, which architectural approach (e.g., cloud, microservices) would be more suitable for scaling the system?
3. In your opinion, what are the most critical aspects of ensuring security and compliance with KVKK/GDPR regulations?
4. For data storage, do you think cloud-based or on-device storage would be more secure and practical?

Interview Results:

The interview revealed that a combination of Flutter and Firebase would be the most efficient and practical solution for SubMind.

Flutter allows the development of both iOS and Android applications from a single codebase, while also offering web compatibility for future expansion.

On the backend side, Firebase's Backend-as-a-Service (BaaS) model was recommended due to its strong integration with Flutter and its ability to automatically scale horizontally based on user volume and database size. This means Firebase can handle increased demand without manual server management, which is highly advantageous for MVP development.

From a security perspective, it was suggested to use TLS encryption and encryption at rest to protect data both in transit and at rest. Firebase Rules can be applied to establish role-based access control, ensuring that only authorized users can access specific data.

The interviewee also highlighted the importance of obtaining explicit user consent and fulfilling data deletion requests in compliance with KVKK and GDPR regulations.

Regarding data storage, cloud-based storage (Firestore) was identified as the most practical and secure solution. Cloud storage enables users to access their accounts from multiple devices, while maintaining strong protection through Firebase's built-in security rules.

It was also emphasized that SubMind should not store payment information directly, as App Store and Google Play already handle payment processing, thereby reducing legal and security risks.

In conclusion, the technical expert recommended that SubMind should be developed on a secure, cloud-based, and easily scalable infrastructure, as this approach would provide both flexibility and long-term sustainability.

Survey

During the preliminary research phase of the SubMind project, a survey was conducted to understand users' subscription management habits, the challenges they face, and their expectations from such a system.

The survey aimed to measure users' current number of subscriptions, their payment tracking methods, the problems they encounter, and their interest in potential features.

Distributed online via Google Forms, the survey reached over 100 participants, primarily university students and young professionals.

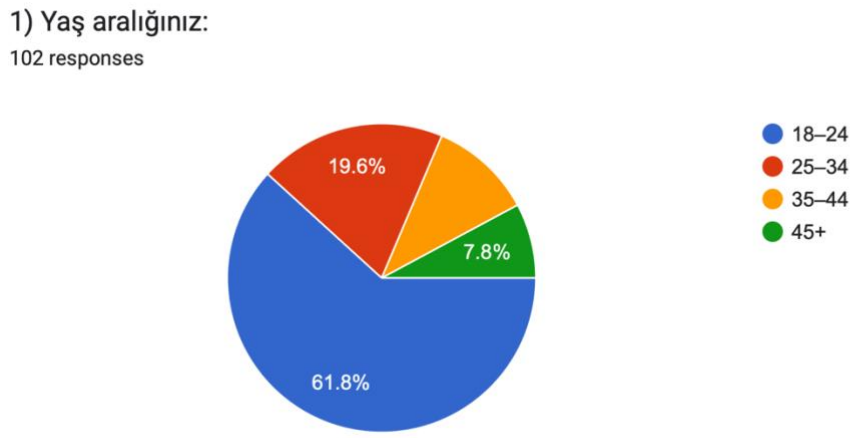
The collected data revealed the main issues users experience in managing their subscriptions and provided valuable insights into which features SubMind should prioritize in its development process.

Survey Link: **https://forms.gle/iwPw2SJQoskwGAGr7**

Survey Results:

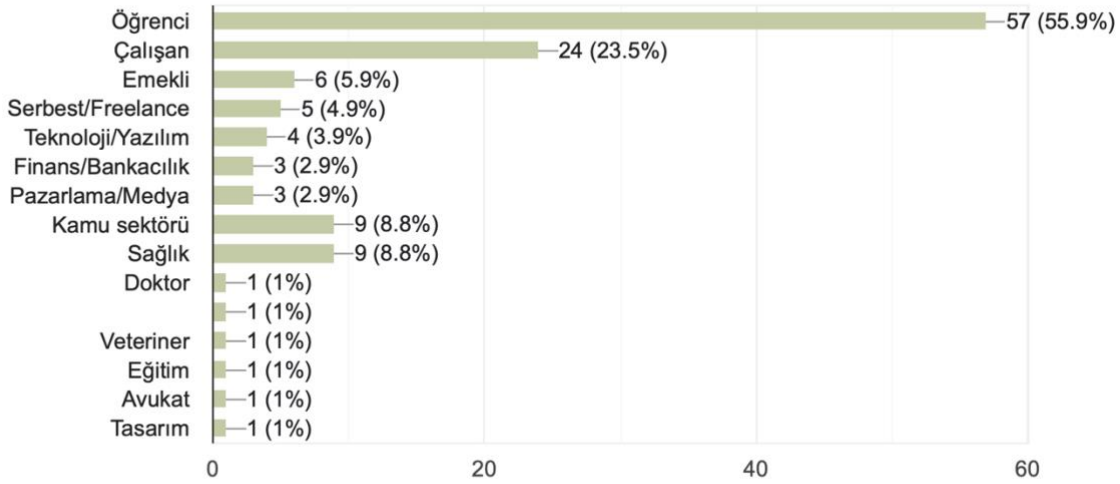
1-Demographic Information:

This section examines the participants' age, occupation, and gender distribution. Data collected from 102 respondents provides valuable insights into the target audience of SubMind.



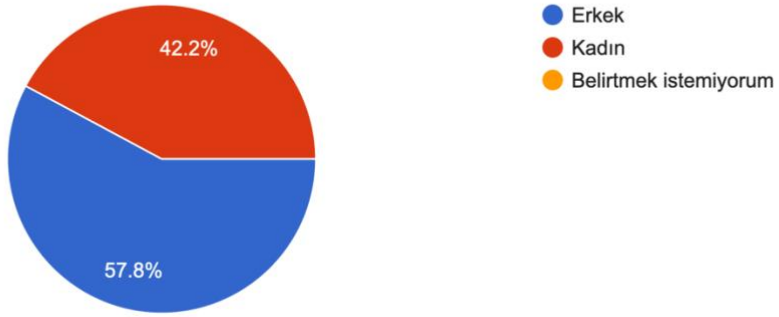
2) İş alanınız / durumunuz (birden fazla seçilebilir):

102 responses



3) Cinsiyetiniz:

102 responses



Findings Summary:

• 61.8% of participants are between the ages of 18–24, indicating that the majority are young adults.

• 55.9% are students, while 23.5% are working professionals, showing a strong representation of young, digitally active individuals.

• The gender distribution is 57.8% male and 42.2% female, suggesting a relatively balanced yet slightly male-dominant user base.

• Additionally, small segments include participants from the public sector (8.8%), technology/software (3.9%), and finance/banking (2.9%), highlighting a diverse occupational background.

Interpretation / Insight:

This demographic profile clearly indicates that SubMind's primary audience consists of students and young professionals.

This group typically manages multiple digital subscriptions (Netflix, Spotify, Amazon Prime, etc.) while also being highly aware of their budgeting and financial habits.

Given their familiarity with technology and preference for mobile solutions, these users are likely to value features that simplify financial management and improve subscription awareness.

Actionable Recommendations:

• The app design should be simple, modern, and youth-oriented, appealing to students and young professionals.

• Targeted marketing should focus on digital channels such as Instagram and TikTok, where this demographic is most active.

• Introduce special plans like "Student Mode" or "Budget Saver Plan" to attract cost-conscious users and strengthen loyalty.

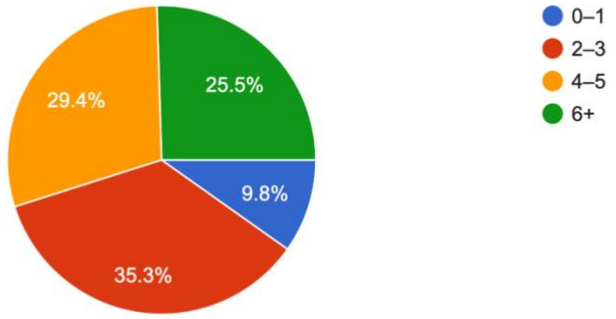
• For working professionals, consider adding expense reports and billing summaries to enhance practicality.

2- Subscription Profile:

This section explores users' subscription behaviors, including the number and types of digital subscriptions they hold, as well as their average monthly spending. The findings provide a detailed understanding of users' digital consumption habits and financial priorities.

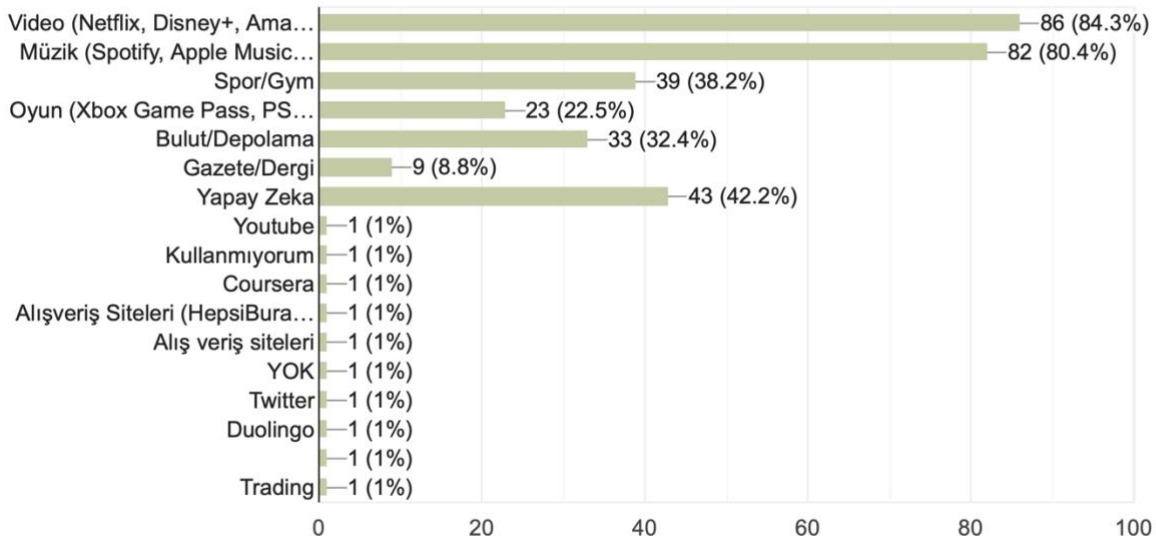
1) Kaç farklı dijital aboneliğiniz var?

102 responses



2) En çok hangi tür abonelikleri kullanıyorsunuz? (birden fazla seçilebilir)

102 responses



Findings Summary:

• Subscription Count: 35.3% of participants have 2–3 subscriptions, 29.4% have 4–5, and 25.5% have 6 or more. Only 9.8% have 0–1, indicating that the vast majority actively manage multiple subscriptions simultaneously.

• Subscription Types: The most common categories are video streaming (84.3%) and music (80.4%), followed by AI tools (42%), fitness/gym memberships (38%), and cloud storage services (32%).

• Monthly Spending: Nearly 46% of users spend more than 1000 TL per month, while 17.6% spend 251–500 TL, and 12.7% spend 0–250 TL. This highlights that subscription expenses represent a significant portion of monthly budgets for many users.

Interpretation / Insight:

The data clearly shows that digital subscriptions are a central part of users' daily lives and financial commitments. A large proportion of users maintain three or more active subscriptions, primarily for entertainment, productivity, and AI-related services.

This pattern reflects both subscription fatigue and a lack of centralized tracking tools users often lose visibility of total costs, renewal dates, and overlapping services.

SubMind can therefore position itself as a comprehensive subscription management solution, offering smart automation, financial insights, and cost optimization particularly valuable for students and young professionals with limited budgets.

Actionable Recommendations:

• Emphasize cost awareness and spending transparency in the app's interface and marketing messages.

• Provide monthly expenditure summaries and trend analytics to help users monitor subscription growth and spending patterns.

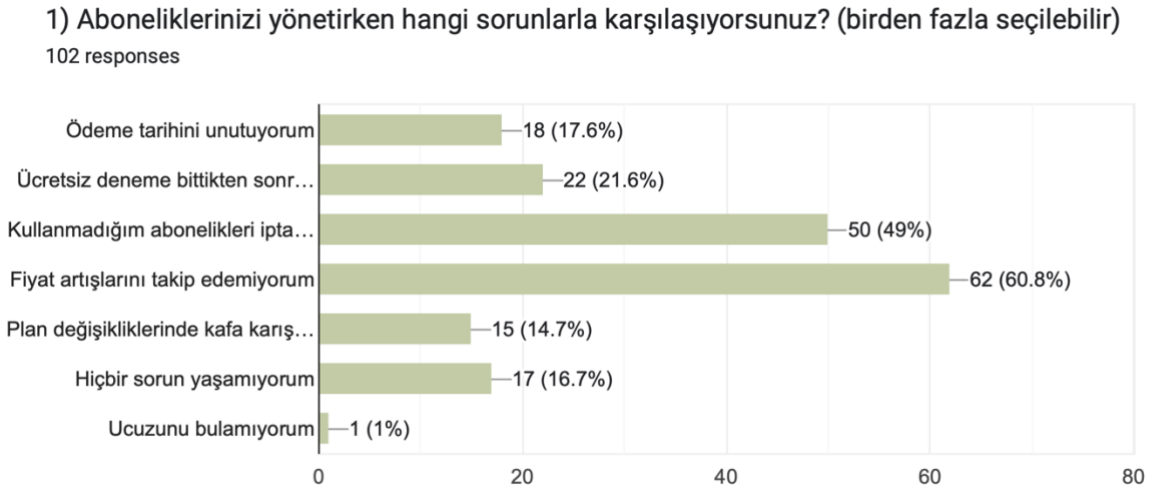
• Enable automatic subscription detection for popular platforms (e.g., Netflix, Spotify, ChatGPT, Adobe, etc.).

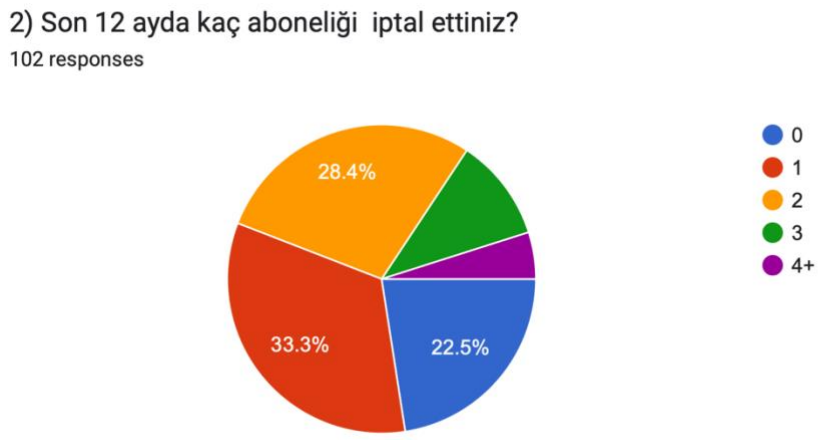
• Add personalized cost-saving features, such as cheaper plan suggestions, inactivity alerts, or free trial expiration reminders.

• Consider gamifying financial awareness (e.g., "monthly saving goals" badges) to increase engagement among younger users.

3- Problems Experienced

This section explores the key difficulties users face while managing their subscriptions, as well as their cancellation habits over the past year. The findings provide valuable insights into the main pain points that SubMind should aim to address in its design and functionality.





Findings Summary:

• The most frequent issue reported was difficulty tracking price increases (60.8%), followed by forgetting to cancel unused subscriptions (49%).

• 21.6% of participants stated that they continued paid subscriptions after a free trial ended, while 17.6% admitted to forgetting payment dates.

• 14.7% experienced confusion over plan changes, and 16.7% reported no significant issues with managing their subscriptions.

• Regarding cancellation behavior, 33.3% of respondents canceled one subscription within the past 12 months, while 28.4% canceled four or more, suggesting that users actively review and adjust their subscriptions throughout the year.

Interpretation / Insight:

The findings indicate that price tracking and cost transparency are the most pressing challenges for users.

Most users fail to notice incremental price increases or overlapping subscriptions until significant money has already been spent. This suggests a growing need for tools that automatically detect price changes, notify users of renewals, and summarize total monthly costs.

The high cancellation activity also reveals that users are aware of redundant spending and are willing to take corrective action when provided with the right insights a behavior SubMind can leverage through timely alerts and optimization suggestions.

Actionable Recommendations:

• Implement real-time price tracking and alert notifications for all major subscription services.

• Develop an “Unused Subscription Detector” that analyzes login or usage frequency to identify inactive subscriptions.

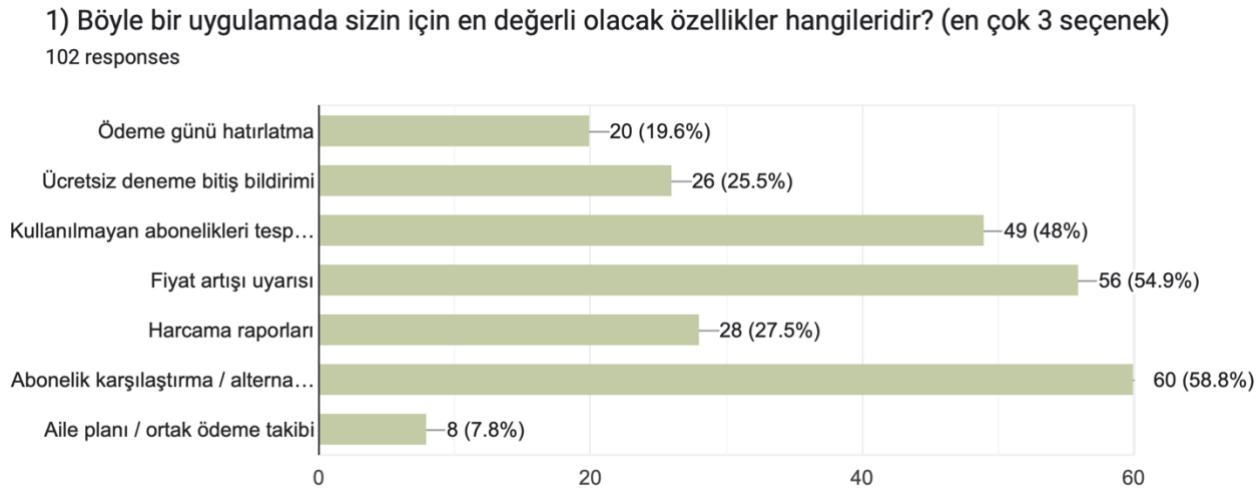
• Add free trial and renewal reminders to prevent unintentional payments.

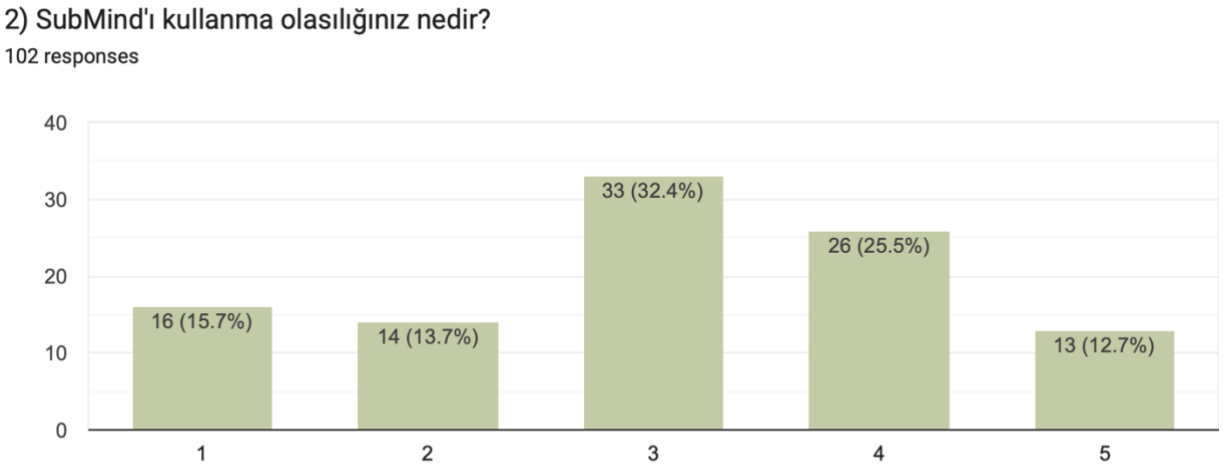
• Create an interactive insights dashboard that highlights cost trends, price increases, and subscription activity.

• Use this insight-driven messaging in marketing campaigns with taglines.

4- Expectations & Desired Features:

This section explores what users expect from a subscription management application like SubMind. The data highlights which features users find most valuable, their likelihood of adopting such a system, and their preferred platform for accessing it.





Findings Summary:

• The most valued feature is subscription comparison and alternative plan suggestions (58.8%), followed closely by price increase alerts (54.9%) and unused subscription detection (48%).

• Other notable features include spending reports (27.5%), free trial expiration notifications (25.5%), and payment reminders (19.6%).

• When asked about adoption likelihood, 32.4% of participants rated their likelihood of using SubMind as 3/5, while 25.5% rated 4/5, indicating moderate to strong interest among potential users.

• In terms of platform preference, 58.8% prefer to access SubMind via a mobile app (iOS/Android), while 41.2% would like it to be available both as a mobile and web application, reflecting users' desire for accessibility across devices.

Interpretation / Insight:

The findings demonstrate that users prioritize financial awareness, optimization, and convenience.

They want an intelligent system that not only tracks but also interprets subscription data, suggesting cheaper alternatives and alerting them to cost changes.

The high demand for mobile access reinforces that SubMind's user base expects on-the-go management — consistent with their dynamic, tech-savvy lifestyle.

Furthermore, the moderate-to-high adoption scores show that there is already a strong market potential if the app provides clear financial value and user-friendly design.

Actionable Recommendations:

• Prioritize key features such as price alerts, alternative plan suggestions, and unused subscription detection in early development stages.

• Develop the system as a mobile-first application, ensuring smooth performance and intuitive UI for iOS and Android.

• Integrate spending visualization dashboards with customizable alerts for renewals, trials, and payment cycles.

• Add AI-driven suggestions for cheaper plans or bundled options to strengthen the app's decision-support role.

• In marketing, emphasize SubMind's smart financial assistant positioning — "Save more, manage smarter."

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*** Note: ChatGPT was used only for linguistic editing and clarity improvement. All written content, ideas, and analyses belong entirely to the project authors.*